

BIT 4107-Mobile Application Development

ENHANCED PROFESSIONAL STUDENT LOGBOOK

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Admission Number:BIT/2024/49929

School/Faculty: COMPUTING AND INFORMATICS

Programme:Blood_Donation_App

Academic Year: THREE

Semester:TWO

Lecturer: MR.MICHAEL NYORO

Phone Number:0725104634

WEEK 1: INTRODUCTION TO MOBILE APPLICATION DEVELOPMENT

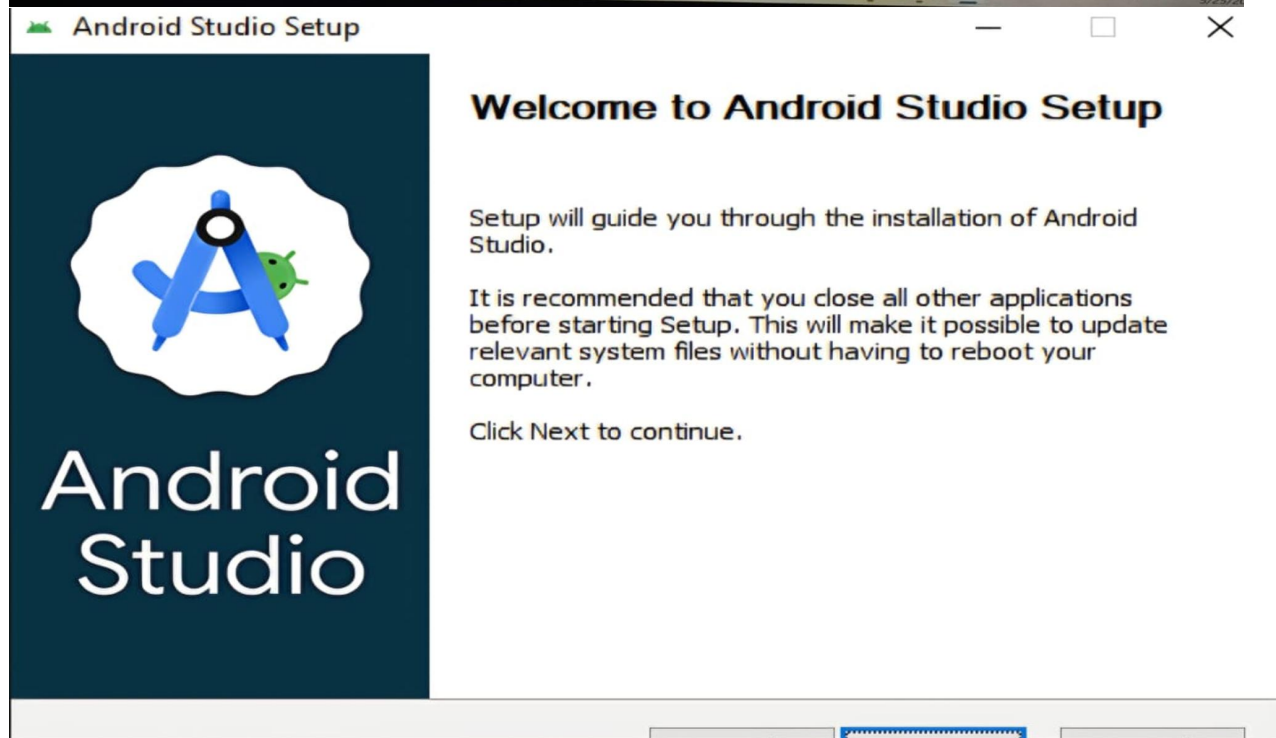
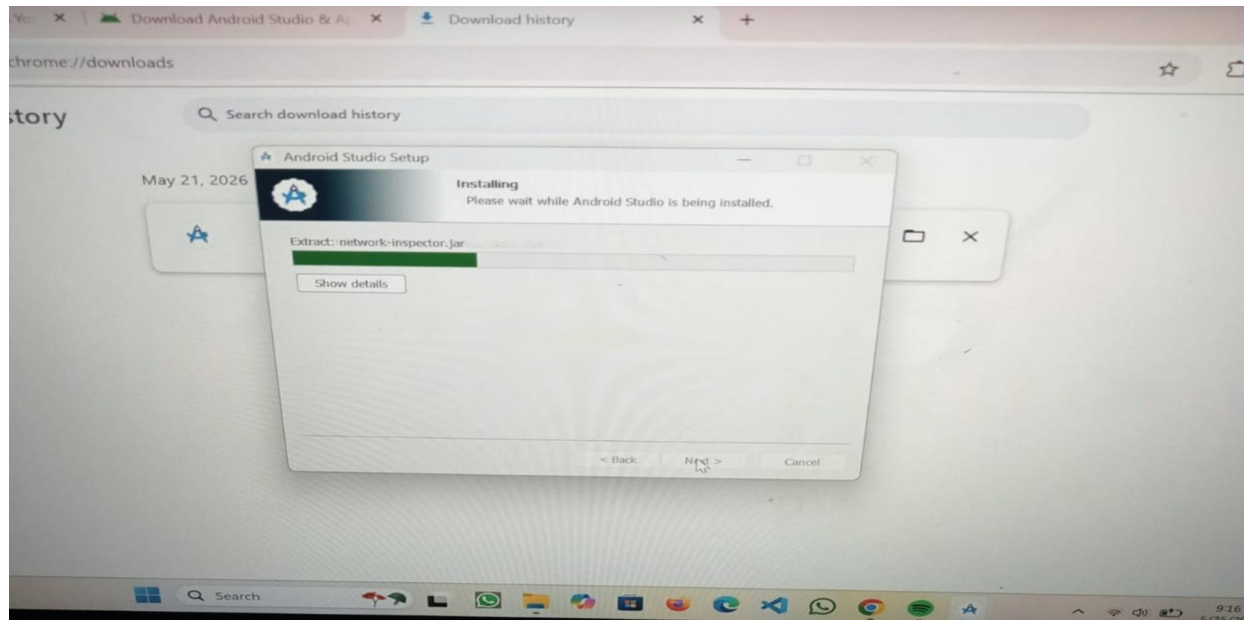
The screenshot displays the Visual Studio Code interface. The top menu bar includes File, Edit, Selection, View, Go, Run, Terminal, and Help. The left sidebar shows the Explorer view with the 'Flutter' extension highlighted. The main editor area displays the 'Flutter' extension details, including the Flutter logo, version v3.42.0, and the Dart Code extension. The bottom panel shows the 'Problems' view with the 'flutter doctor' command output.

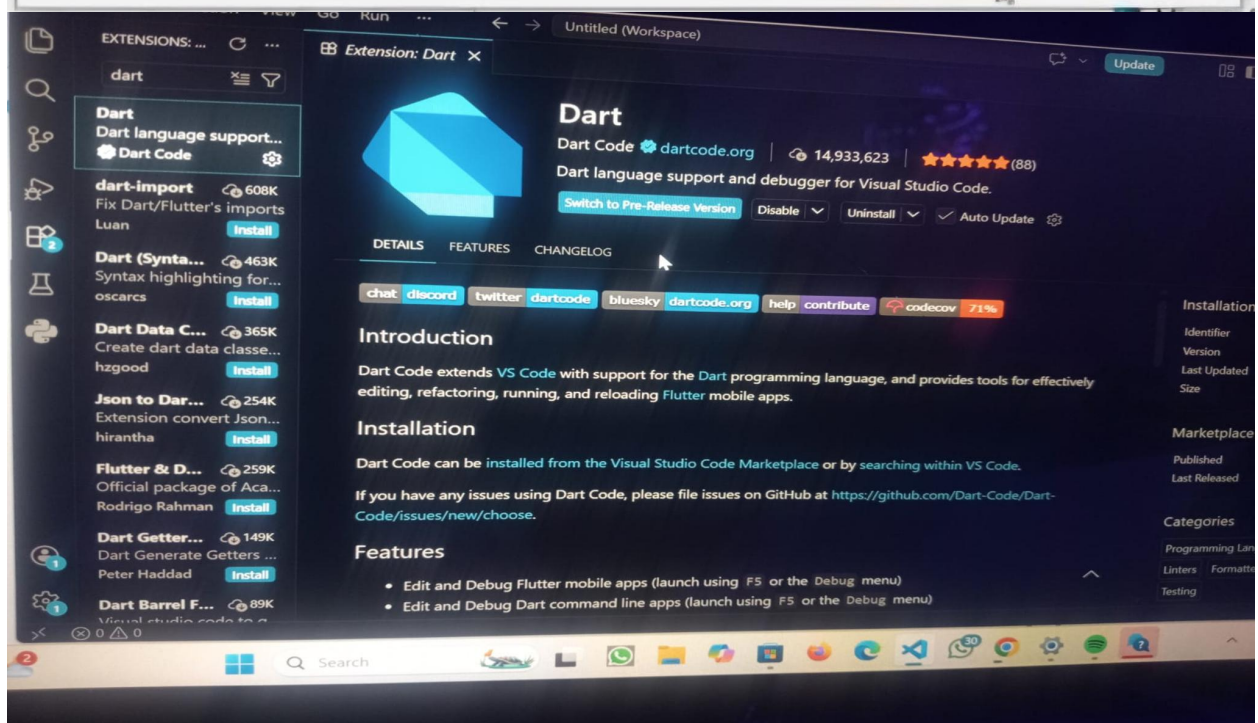
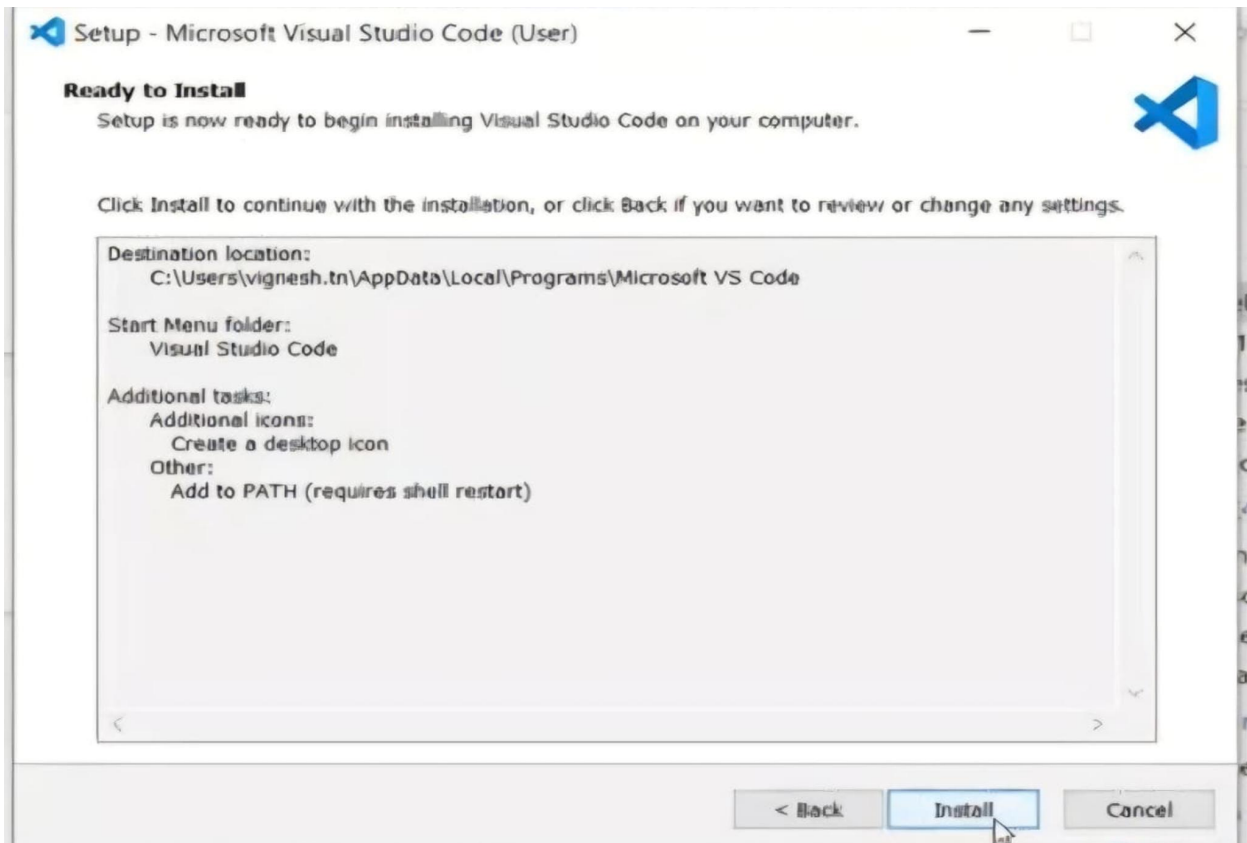
Flutter Extension Details:

- Flutter v3.42.0
- Dart Code | 4,380,818 | ★
- Flutter support and debugger for Vis...
- Disable | Uninstall | ⚙️
- This extension is enabled globally.

flutter doctor Output:

```
PS C:\Users\HP> flutter doctor
Doctor summary (to see all details, run flutter doctor -v):
[✓] Flutter (Channel stable, 3.44.0, on Microsoft Windows [Version 10.0.19045.6456], locale en-US)
[✓] Windows Version (10 Pro 64-bit, 22H2, 2009)
[!] Android toolchain - develop for Android devices (Android SDK version 36.1.0)
    X cmdline-tools component is missing.
      Try installing or updating Android Studio.
      Alternatively, download the tools from https://developer.android.com/studio#command-line-tools-only and make sure to set the ANDROID_HOME
      environment variable.
      See https://developer.android.com/studio/command-line for more details.
    X Android license status unknown.
```





TOOLS AND SOFTWARE USED

1.Android Studio

2.VS Code

3.Flutter SDK

4.Dart

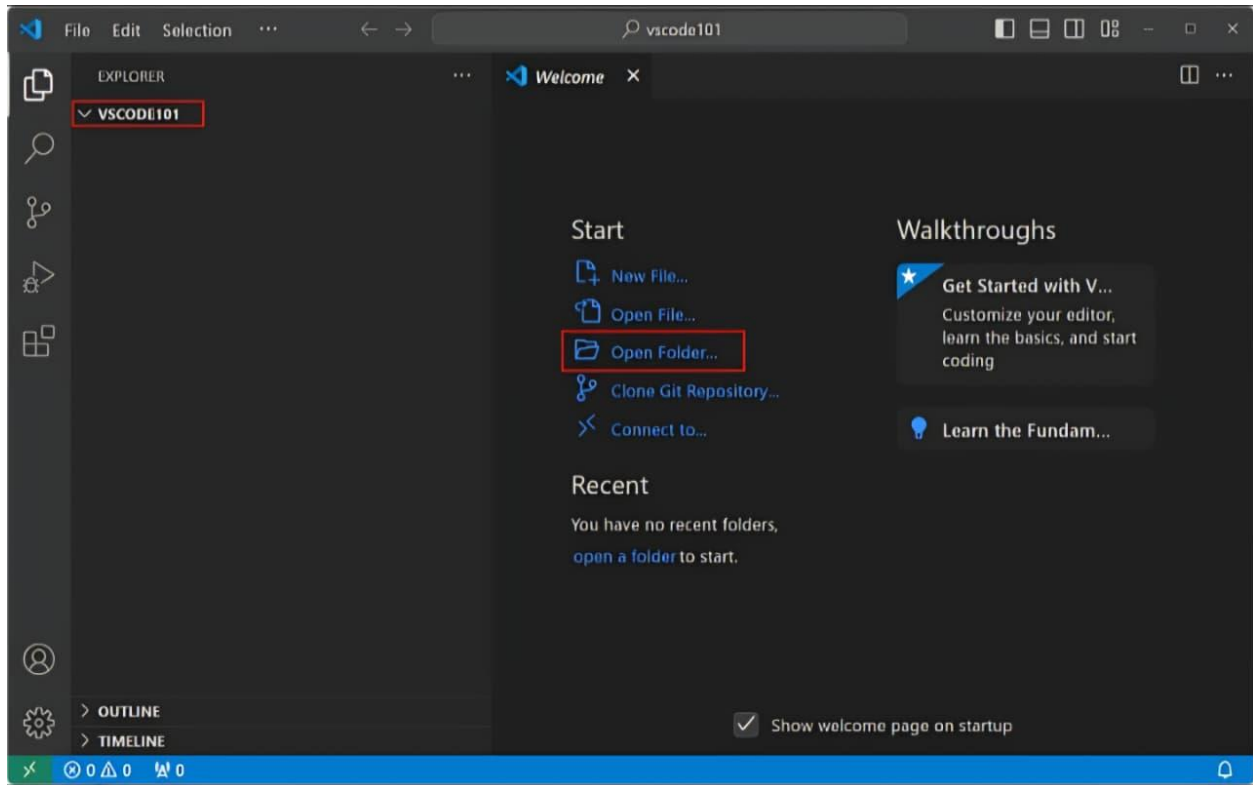
5.Android Phone

PERSONAL REFLECTION

Installing the tools was challenging at first due to errors and configuration issues, but it improved my problem-solving skills. I gained practical experience in setting up development environments and became more confident in using modern software tools for app development.

WEEK 2: MOBILE DEVELOPMENT

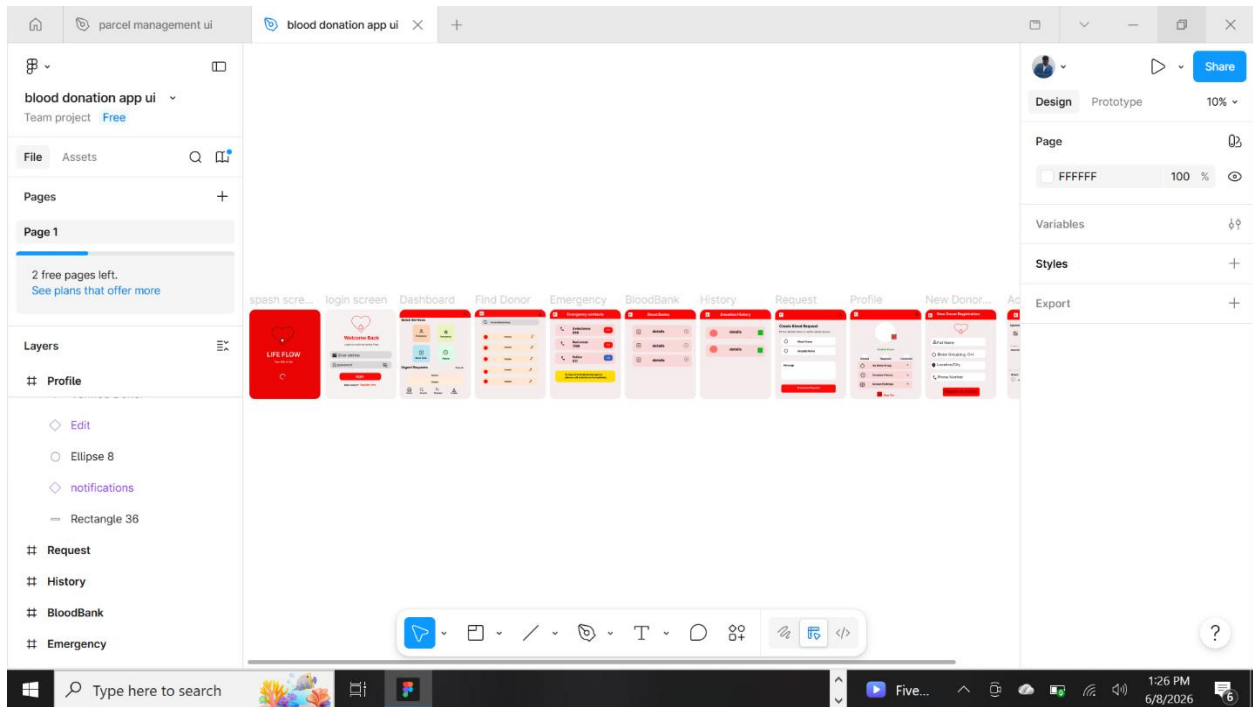
ENVIRONMENT



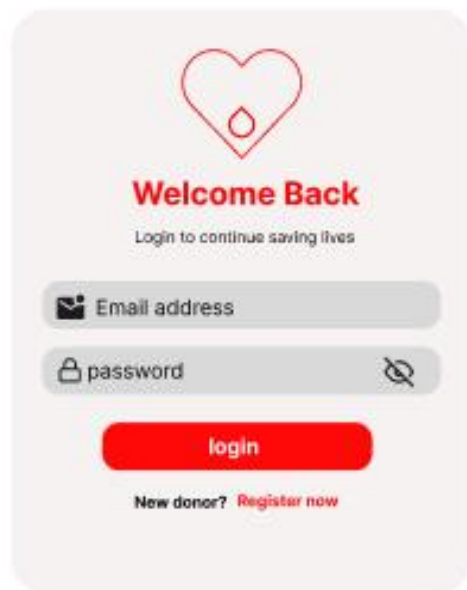
PERSONAL REFLECTION

I set up the mobile development environment by installing and configuring tools like Android Studio, Flutter SDK, and device emulators. This process helped me understand how different components work together, though I faced minor setup challenges that enhanced my troubleshooting and technical skills.

WEEK 3: USER INTERFACE DESIGN



login screen



spash screen



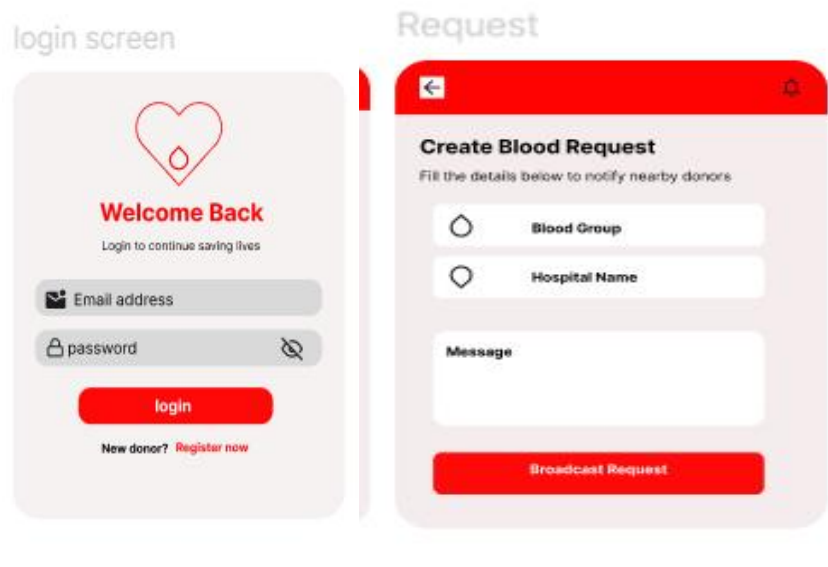
Dashboard



Personal reflection

I focused on user interface design for the blood donation app using Figma. I learned how to create and organize screens, design user-friendly layouts, and improve overall system usability. This improved my understanding of UI/UX principles and enhanced my design skills.

Week 4: Event handling and user interaction



18:54



Account Settings

Appearance



Dark Mode



Change Password

Old Password

New Password

Cancel

Save

Security



About



1.0.0

1 2 3 4 5 6 7 8 9 0

q w e r t y u i o p

a s d f g h j k l

⬆ z x c v b n m ⬇

!#1 , English (UK) . Done



Personal reflection

I learned about event handling and user interaction in the blood donation app. I implemented how users interact with the system through buttons, forms, and input fields. This improved my understanding of how applications respond to user actions and enhanced my programming skills.

Week 5: Data management

19:08



New Donor Registration



Full Name



Kelly maroa

Blood Group (e.g. O+)



O+

Location/City



Machakos

Phone Number



0785324646

Register as Donor



Search Blood Group



Anna Itotiah W.
Nairobi



Sharon Kendi.
Nakuru



Faith Mueni.
Mombasa



Roy Gichinga.
Kisumu



Michelle Njogu.
Eldoret



Kelley Maroa
Migori



Nicole Chepkemboi
Nandi



Home



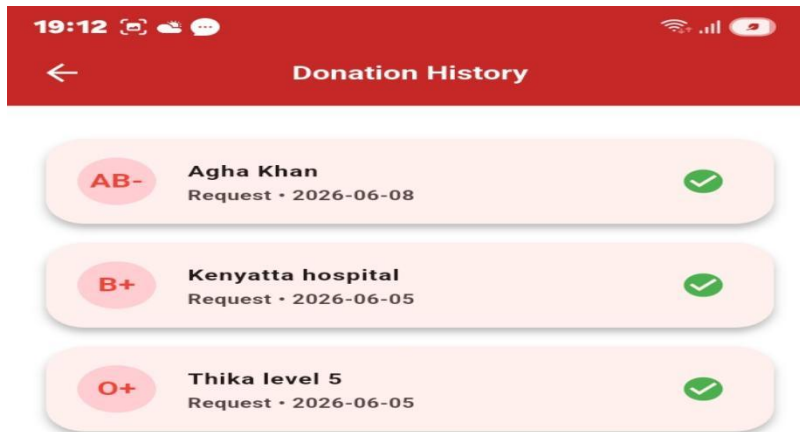
Search



Request



Profile



Personal reflection

I focused on data management in the Blood Donation App. I learned how to store, retrieve, and update donor and request information using local storage and databases. This improved my understanding of how data is handled efficiently and enhanced my skills in managing application data.

Week 6: Networking and APIs

```
1  {
2    "project_info": {
3      "project_number": "596793580149",
4      "project_id": "blood-donation-app-a7042",
5      "storage_bucket": "blood-donation-app-a7042.firebaseioapp"
6    },
7    "client": [
8      {
9        "client_info": {
10          "mobilesdk_app_id": "1:596793580149:android:d5f7eac4ef37ed8144363c",
11          "android_client_info": {
12            "package_name": "com.blood_donation.app"
13          }
14        },
15        "oauth_client": [],
16        "api_key": [
17          {
18            "current_key": "AIzaSyDVDy62fZd7rrAi_DWkWh7gyK9Nb91T9Wo"
19          }
20        ],
21        "services": {
22          "appinvite_service": {
23            "other_platform_oauth_client": []
24          }
25        }
26      }
27    ],
28    "configuration_version": "1"
29  }
```

A blood donation app requires networking to connect users with donors and hospitals, enable real-time updates, send requests, share location data, and store information on cloud servers for accessibility and communication.

